Diagnosing Goals in Soccer: Breakdown vs Breakthrough

Tioluwani Akinloye
Simon Fraser University, CMPT 742

1 Introduction

Understanding why a goal was conceded is crucial for coaching and tactical analysis, yet analysts still judge by eye whether the resulting goal was a result of out of shape defense, which frankly is slow, subjective, and hard to scale. Tt seems that existing studies focus more on whole-team synergy rather than defensive success specifically, and even more so rarely over the analysis of a single goal.

The video-based reconstruction that is commonly done, semi-exclusively produces player positions but stops before attempting any tactical interpretation. I'm proposing a system that takes a video clip of the build-up to a goal and flags whether the defense lacked integrity (specifically defined). It *displays* its evidence as defensive radii and inter-defender lines, with a range of intensity gradients overlaid on the video (Figure **??**), so a single conceded goal can be diagnosed without needing tracking data on individual players.

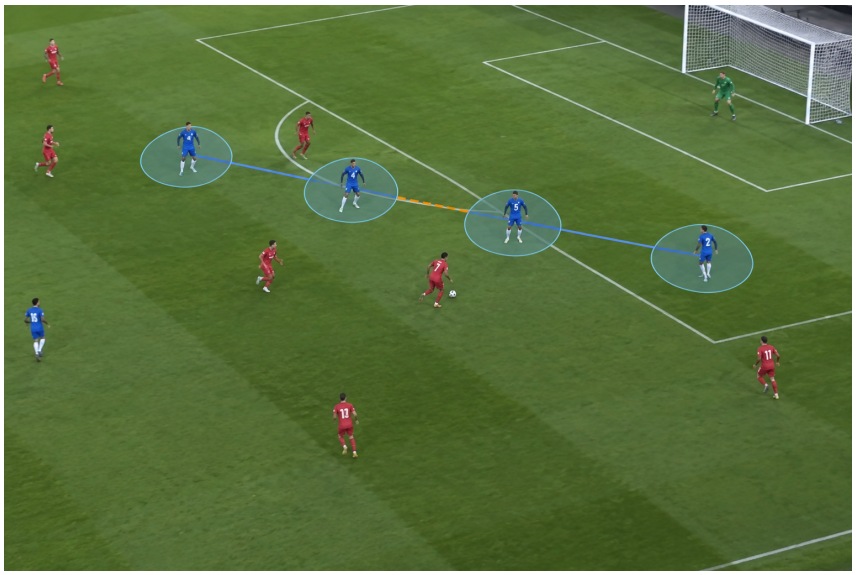


Figure 1: A frame of a clip, where defender radii is applied and there are lines in-between defenders, later resulting in the goal being labeled with it's faults (AI generated image)

2 Definitions

A goal is a *breakdown goal* if it is the result a structural breakdown, meaning at least one formational/positional fault (broken lines, large gaps, defenders out of position) occurs within an arbitrary window before the shot. Otherwise it is a *breakthrough goal*, one scored in spite of a defense that held its shape. These labels are operational definitions: what I mean by that is they simply describe measured geometry, not necessarily the ground truth cause of the goal or the quality of either team, (all thresholds will likely be tunable parameters). I deliberately restricted faults to formation and positioning and exclude one-on-one defending, goalkeeping, and other individual factors, so a breakthrough goal only means that no structural fault was detected, this was done as per your advice on avoiding subjective definitions.

3 Proposal

Input. A 10–15 s clip ending with a goal (likely from a consistent bird eye-sh view, to avoid dealing with tracking consistency over camera changes).

Output. (i) The video with radii and defender lines overlaid, (ii) the defined label (along with the specific triggering faults if it was a breakdown goal) (iii) a top-down tactical pitch view with dots instead of players matching the gameplay (this is a maybe) (iv) a short feature summary

Metrics. Each defender's *defensive radius* method TBD, (fixed distance first, then speed and direction based possibly) and then *inter-defender line*, where line gaps, line height, compactness, and defenders goal-side of the ball will be considered (Figure 2).

Tech stack and data. (Prospective) The pipeline would likely written in Python with PyTorch and NumPy, maybe YOLO11 for player and ball detection, BoT-SORT for tracking, OpenCV for overlays. For data, I'm going to start with SoccerNet and Metrica.

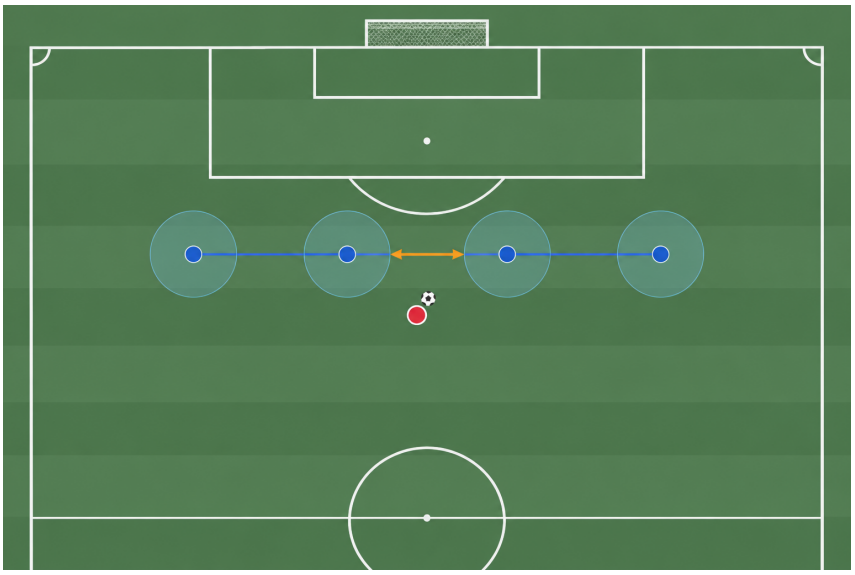


Figure 2: Pressure metrics (AI-generated image).

3.1 'Triggering Faults'

What's kind of in my head for the triggering faults that determine a structural breakdown. Let the critical window be the final T seconds before the shot ($T = \textit{some value}$). Table 1 lists the faults; values are to be tuned.

Fault	**Condition (team-level geometry)**	**Values**
Line gap	Gap between adjacent back-line defenders exceeds τ_g.	$\tau_g = ...$m
Lagging defender	A defender is more than τ_l out of line depth.	$\tau_l = ...$m
Delayed shift	Line center lags the ball laterally by more than τ_s for over 1 s.	$\tau_s = ...$m
Numerical deficit	Fewer than τ_n defenders goal-side of the ball.	$\tau_n = ...$
Unpressured ball	No defender within radius r of the ball for $\tau_.$..s.	$r = ...$ m, $\tau_p = f$s

Table 1: Structural faults (purely geometrical.)

4 Timeline

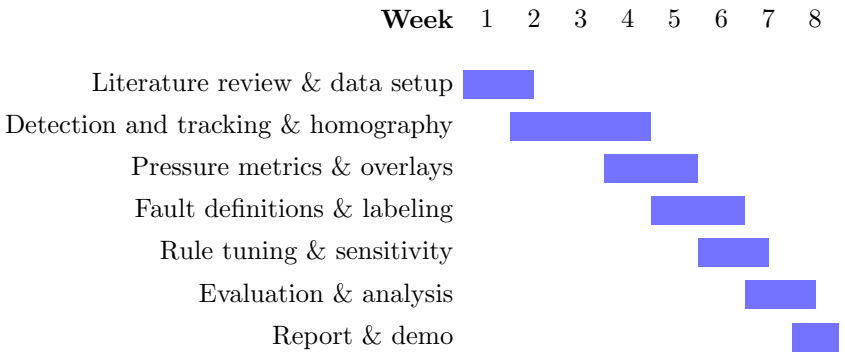


Figure 3: Project timeline over eight weeks (give or take).

Weeks 1–4 should cover most of the preambulatory work: setup, data, video pipeline;
Weeks 4–7 prioritize metrics, fault rules, labeling, and fine-tuning; and then;
Weeks 7–8 for evaluation, the presentation, and a demo.

Core Deliverable: My goal will be to have released code and annotated videos for at least 25 goals (within a fairly diversified set of matches).

References

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[2] Dash, S., Ide, K., Umemoto, R., Amino, K., Fujii, K.: Prediction-based evaluation of back-four defense with spatial control in soccer. arXiv:2511.06191 (2025)

[3] Bischofberger, J., Ma, R., Bauer, P., Arnsmeyer, K., Baca, A.: Blame is easier than praise: Measuring off-ball defensive performance in football. arXiv:2606.19931 (2026)

[4] Mhmdwael77: ai-football-analytics: End-to-end football computer-vision analytics with YOLO11 and BoT-SORT. GitHub repository. https://github.com/Mhmdwael77/ai-football-analytics